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A Closed-Form r=2 Prior for Mantissa Bits and Why Seeding an Adaptive Coder With It Yields No Practical Gain

Evidence-grounded FCN academic paper

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Abstract

We report a closed-form expression for the per-bit bias of BF16 mantissa bits in trained neural network weights and an empirical evaluation of using that expression as a seed prior for an adaptive arithmetic coder. The closed form follows Benford's law applied to binary floating-point significands

and gives a per-bit bias $\mathrm{bias}_i = B_0 \cdot 2^{i-M+1}$ with $B_0 = -(\log_2 3 - 3/2) \approx -0.08496$ for the most significant mantissa bit, decaying toward zero for the least significant bits. The formula correctly predicts the empirically observed per-bit biases on BF16 weights of a publicly released large language model. However, when we instantiate the formula as a seed prior for an adaptive arithmetic coder and compare against an unseeded adaptive coder on the same input, we observe no net compression benefit. The seeded coder attains a reported compression ratio of approximately $1.522\times$, indistinguishable from the unseeded baseline within the precision reported. We interpret the result as evidence that real data overwhelms the prior within the first roughly one thousand values per tensor, so the closed-form prior, although correct, provides no practical advantage for coding under the tested conditions. The claim is scoped to the tested setting and does not extend to cross-model conditioning, learned non-IID coders, format redesign, or task-specific model compression unless explicitly stated.

Keywords: lossless compression; neural network weights; Forlais compression research; BF16; mantissa bit bias; Benford's law; arithmetic coding; adaptive coder; prior seeding

Methodology Disclosure Boundary

This paper states the FCN evidence for the reported result at a public-safe abstraction level. It reports measured quantities, reported scope, and limitations. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, model weights, infrastructure, or operational procedures.

1 Introduction

Lossless compression of trained neural network weights is studied across the FCN research programme as a coding question first and an engineering question second. For each FCN entry, the goal is to state the smallest public-safe claim that the evidence actually supports, with the format, scope, and reconstruction target made explicit.

This paper reports the FCN evidence for two distinct, related results. First, a closed-form expression that predicts the per-bit bias of BF16 mantissa bits in trained weights, derived from Benford's law applied to binary floating-point significands. Second, an empirical evaluation of using that closed-form expression as a seed prior for an adaptive arithmetic coder, compared against the same adaptive coder initialised with a uniform prior. The two results are reported together because they illuminate each other: the formula is correct, but its translation into a coder benefit is not automatic, and the absence of that benefit is itself instructive.

The headline numbers, sourced from the FCN evidence register, are: the closed-form constant $B_0 \approx -0.08496$ for the most significant mantissa bit, with the full sequence $\mathrm{bias}_i = B_0 \cdot 2^{i-M+1}$ for $i = 0, \ldots, M-1$ where $M = 7$ for BF16; a reported compression ratio of approximately $1.522\times$ on the tested setting for the seeded adaptive arithmetic coder; and the same approximately $1.522\times$ for the unseeded baseline within the precision reported. The seed provides no measurable coding benefit.

The remainder of the paper states the public-safe scope, summarises the relevant background, derives the closed-form prior, presents the verification protocol and the empirical comparison, discusses why the correct prior yields no practical coding benefit, and records the limitations and disclosure boundary.

2 Scope

The result reported in this paper concerns:

- BF16 stored weights of a publicly released large language model;
- a closed-form prior over per-bit biases of the BF16 mantissa, derived from Benford's law applied to binary floating-point significands;
- a single-tensor adaptive arithmetic coder seeded with that closed-form prior, compared against the same coder seeded with a uniform prior;
- single-model lossless compression with bit-exact reconstruction verified by SHA-256 round-trip;
- public-safe summarisation of measured quantities.

The result does not concern:

- lossy model compression, pruning, distillation, and quantisation-aware retraining as separate regimes;

- cross-model delta coding against a paired base model, which is reported in separate FCN entries;
- learned non-IID coders trained specifically on weight distributions;
- format redesign or architecture-specific compression that requires modifying model semantics;
- content-addressable deduplication, which is reported in separate FCN entries.

Each excluded regime is a distinct setting with its own evidence base. Ratios from those settings should not be compared with the headline number in this paper without first reconciling assumptions about coding scope, format, and reconstruction target.

3 Background and related work

Lossless compression of trained neural network weights sits at the intersection of classical source coding and contemporary neural network engineering. Shannon's source coding theorem gives a lower bound on the expected code length of any lossless code in terms of the entropy of the source distribution [17, 2]. Arithmetic coding approaches that bound to within an additive constant for any distribution that can be modelled adaptively, with practical realisations attributed to Rissanen and to Witten, Neal, and Cleary [14, 18, 16]. Asymmetric numeral systems are a related entropy-coding family [3], and Krichevsky-Trofimov-style universal coders provide a classical reference for the convergence of adaptive coders from a prior to the empirical distribution [9].

Benford's law describes the distribution of significant digits in many empirical data sources, including outputs of multiplicative processes [12, 1, 13, 4]. Trained-weight distributions in neural networks are shaped in part by stochastic-gradient training [15, 8], which contains multiplicative dynamics; this places trained weights within the class of distributions to which Benford's law applies.

Modern model weights are typically stored in BF16 or FP32. The relevant numeric formats are specified in the IEEE 754 standard and discussed in the BF16 and mixed-precision training literature [5, 7, 10]. We refer to the publicly released model artefacts associated with the architecture examined here [6].

The companion FCN entry that derives Benford's law as the origin of the BF16 entropy ceiling provides the parent context for this paper [12, 1]; the present paper derives a coder-side consequence of that derivation and reports its empirical effect.

4 Theoretical analysis

4.1 Setup

BF16 stores a real value as a sign bit, an 8-bit exponent, and a 7-bit mantissa, with an implicit leading 1 in the significand. Let $M = 7$ denote the mantissa width and let $i \in \{0, \dots, M-1\}$ index mantissa bits, with $i = M-1$ being the most significant mantissa bit (MSB) and $i = 0$ the least significant. For a given exponent, the significand $s = 1 + m/2^M$ for an integer mantissa value $m \in \{0, 1, \dots, 2^M - 1\}$.

4.2 Benford-derived mantissa distribution

Benford's law states that, for a real-valued multiplicative process, the probability that the leading significand lies between s and $s + ds$ is proportional to $1/s$ [1, 4]. Applied to the BF16 significand, the probability that the integer mantissa value equals m is

$$\Pr(m) \;=\; \log_2\!\left(\frac{2^M+m+1}{2^M+m}\right), \tag{1}$$

which is the mass of the interval $[s_m, s_{m+1}) = [1 + m/2^M, 1 + (m+1)/2^M)$ under a $\log_2$ scale on the significand.

4.3 Closed form for the MSB bias

The mass of m values with MSB equal to 1 is the sum of equation (1) over the upper half of the mantissa range. Telescoping the logarithm yields

$$\Pr(\mathrm{MSB}=1) \;=\; \sum_{m=2^{M-1}}^{2^M-1} \log_2\!\left(\frac{2^M+m+1}{2^M+m}\right) \;=\; \log_2\!\left(\frac{2^{M+1}}{2^M+2^{M-1}}\right) \;=\; \log_2\frac{4}{3}. \tag{2}$$

The corresponding MSB bias relative to the uniform reference of $1/2$ is

$$B_0 \;\equiv\; \Pr(\mathrm{MSB}=1) - \frac{1}{2} \;=\; \log_2\frac{4}{3} - \frac{1}{2} \;=\; -\left(\log_2 3 - \frac{3}{2}\right) \;\approx\; -0.08496. \tag{3}$$

The MSB is therefore biased toward 0 relative to the uniform reference by approximately 0.085.

4.4 Closed form for less significant mantissa bits

For bit i less significant than the MSB, the Benford-induced bias relative to the uniform reference scales geometrically with the bit's contribution to the significand. To first order, $\mathrm{bias}_i \approx B_0 \cdot 2^{i-M+1}$, so $\mathrm{bias}_{M-1} = B_0$ and the biases decay by a factor of two per bit toward bias_0, which is essentially zero. The full sequence for BF16 ($M = 7$) is therefore approximately $\{B_0, B_0/2, B_0/4, \ldots, B_0/64\}$, decaying from the MSB toward the LSB. This sequence is the closed-form prior that the present paper evaluates.

4.5 Implied entropy and ceiling

The companion FCN entry that derives Benford's law as the origin of the entropy ceiling combines the closed-form mantissa distribution with the measured exponent and sign entropies to obtain a per-value entropy of approximately 10.51 bits per value, and an associated single-model compression ceiling of approximately $1.522\times$ for BF16. The present paper does not re-derive the ceiling; it evaluates whether the closed-form prior translates into a coder-side improvement.

5 Verification protocol

The verification protocol that underlies the empirical comparison is summarised below; lower-level implementation details remain internal.

- Input: BF16 tensors of a publicly released large language model, loaded directly from the published checkpoint format.
- Coder: a single-tensor adaptive arithmetic coder that models each mantissa bit position with its own per-bit binary context, updated online with a standard Krichevsky-Trofimov-style estimator [9].

- Conditions: the coder is run twice on the same input, once initialised with the closed-form Benford prior $\mathrm{bias}_i = B_0 \cdot 2^{i-M+1}$ and once initialised with a uniform prior of $1/2$ for every bit position.
- Acceptance criterion: a run is counted as successful only when the decoded byte stream reconstructs the original BF16 tensor exactly. Reconstruction is verified by SHA-256 round-trip against the original tensor bytes [11].
- Reported ratios: total raw bytes divided by total compressed bytes, aggregated across the tested tensors of the model and reported at the public-safe abstraction level. The seeded and unseeded ratios are reported on the same input.

The protocol is deliberately minimal: only the prior changes between the two conditions, so any observed difference in compressed size is attributable to the prior.

6 Empirical results

Reported quantity	Public-safe value
Closed-form constant B_0	approximately -0.08496
Closed-form per-bit bias formula	$\mathrm{bias}_i = B_0 \cdot 2^{i-M+1}$, $M = 7$
Seeded adaptive arithmetic coder ratio	approximately 1.522x
Unseeded adaptive arithmetic coder ratio	approximately 1.522x
Net coding benefit of the closed-form seed	approximately 0 within reported precision
Approximate values per tensor before prior is overwhelmed	order 10^3
Single-model entropy ceiling (companion FCN entry)	approximately 1.522x
Reconstruction	bit-exact, SHA-256 verified
Numeric format	BF16
Internal measurement date	2026-02-15

Table 1: Summary of public-safe FCN evidence for the closed-form Benford prior and the seeded versus unseeded adaptive arithmetic coder comparison reported in this paper.

The seeded and unseeded ratios are indistinguishable within the precision reported here. Diagnostic measurements on the per-bit empirical distributions match the closed-form prior to high accuracy on the tensors examined, so the absence of a coder-side benefit cannot be attributed to a wrong formula. The next subsection explains why a correct prior nevertheless yields no measurable gain.

6.1 Why the correct prior provides no coding benefit

An adaptive arithmetic coder with a Krichevsky-Trofimov-style update converges from any reasonable prior to the empirical distribution at a rate $O(\log n/n)$ in code length per symbol, where n is the number of observed symbols [9]. The total redundancy from the prior is therefore bounded by a logarithmic term in n. On tensors with millions of values per bit position, this logarithmic term is amortised away to a fraction of a bit per bit position over the whole tensor, which is negligible relative to the entropy ceiling.

Concretely, after on the order of 10^3 values per bit position, the empirical frequency has been updated enough to overwhelm any reasonable initial prior, so the seeded and unseeded coders produce essentially identical code streams from that point onward. The total compressed length differs only by a small constant determined by the first few hundred symbols, which is invisible at the precision at which we report aggregate ratios.

6.2 What this rules out and what it leaves open

The empirical comparison rules out one specific direction: using the closed-form Benford prior as a fixed seed for an adaptive arithmetic coder does not improve compression on the tested setting. It does not rule out other uses of the formula. In particular, the formula may still be useful as a sanity check on observed per-bit biases, as a diagnostic for non-Benford tensors such as normalisation-layer weights, as a structural input to non-adaptive coders that do not learn from data, or as a starting point for closed-form codecs whose code lengths do not depend on a per-tensor model fit. We do not evaluate those uses here; we report only that the most direct translation of the formula into a coder-side seed yields no measurable benefit.

7 Connection to public literature

The result in this paper is consistent with the classical convergence theory for adaptive entropy coders [9, 18, 14]: adaptive coders converge to the empirical distribution at a rate that absorbs any reasonable initial prior within a small number of symbols. The result also corroborates the Benford-derived view of the BF16 entropy ceiling [1, 4]: the formula correctly describes the source, but the source already speaks loudly enough through any reasonable adaptive coder.

8 Implications

For practitioners considering whether to invest in a closed-form prior for an adaptive coder on BF16 weights, the practical implication is that the investment does not pay off at the aggregate level. The implication is not that the formula is wrong; the implication is that the coder absorbs the prior quickly. Improvements in this regime require leaving the regime: cross-model conditioning, format change, lossy regimes, or specialised non-IID coders. We address each of these regimes in separate FCN entries.

9 Discussion

The pattern in this paper is a useful one to surface explicitly: a closed-form prediction can be exactly right about a source and still translate into no measurable improvement at the coder. The reason in this case is the standard convergence rate of adaptive entropy coders. We report the negative result rather than discard it because it constrains the design space: any future codec that claims to extract additional single-model compression from BF16 weights by exploiting mantissa-bit bias alone has to explain why it succeeds where the seeded adaptive coder does not. We do not claim that the formula is useless; we claim only that its most direct coder-side instantiation yields no measurable benefit on the tested setting.

10 Limitations

We note four limitations.

1. The reported ratios are empirical and apply to the specified architecture, numeric format, and tensor role mix; we do not claim universality beyond the tested networks.
2. The result is restricted to lossless, bit-exact reconstruction verified by SHA-256 round-trip. Lossy regimes are out of scope here.
3. The closed form is derived under the Benford-process assumption for trained weights; it does not apply unmodified to normalisation-layer weights or to weights with markedly non-Benford exponent distributions, which are reported in separate FCN entries.
4. Public reproducibility is limited by disclosure control; internal verification artefacts are maintained by Forlais Group but are not included in this paper.

11 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here.

12 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

13 Conclusion

The FCN evidence supports two scoped conclusions. First, a closed-form Benford-derived prior for the per-bit bias of BF16 mantissa bits in trained weights of a publicly released large language model is given by $\mathrm{bias}_i = B_0 \cdot 2^{i-M+1}$ with $B_0 = -(\log_2 3 - 3/2) \approx -0.08496$ and $M = 7$, and matches the empirically observed per-bit biases. Second, using that closed-form prior as a seed for an adaptive arithmetic coder yields no measurable compression benefit relative to a uniform prior on the tested setting; both attain a reported compression ratio of approximately $1.522\times$, consistent with the FCN single-model entropy ceiling. The result is reported here in its public-safe form and should be read alongside the limitations enumerated above.

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Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN academic paper package generated from source evidence.	Oliver Cooper
0.2	14 May 2026	Expanded references, structured evidence summary, scope, and limitations.	Oliver Cooper
0.3	14 May 2026	Full evidence-grounded manuscript with public-safe abstraction, table summaries, and category-aware framing.	Oliver Cooper
1.0	14 May 2026	Publication-candidate manuscript: closed-form derivation of the Benford-induced per-bit prior, seeded versus unseeded coder comparison, convergence-rate explanation of the null result, full limitations.	Oliver Cooper